

RASE Sensor Control Interface

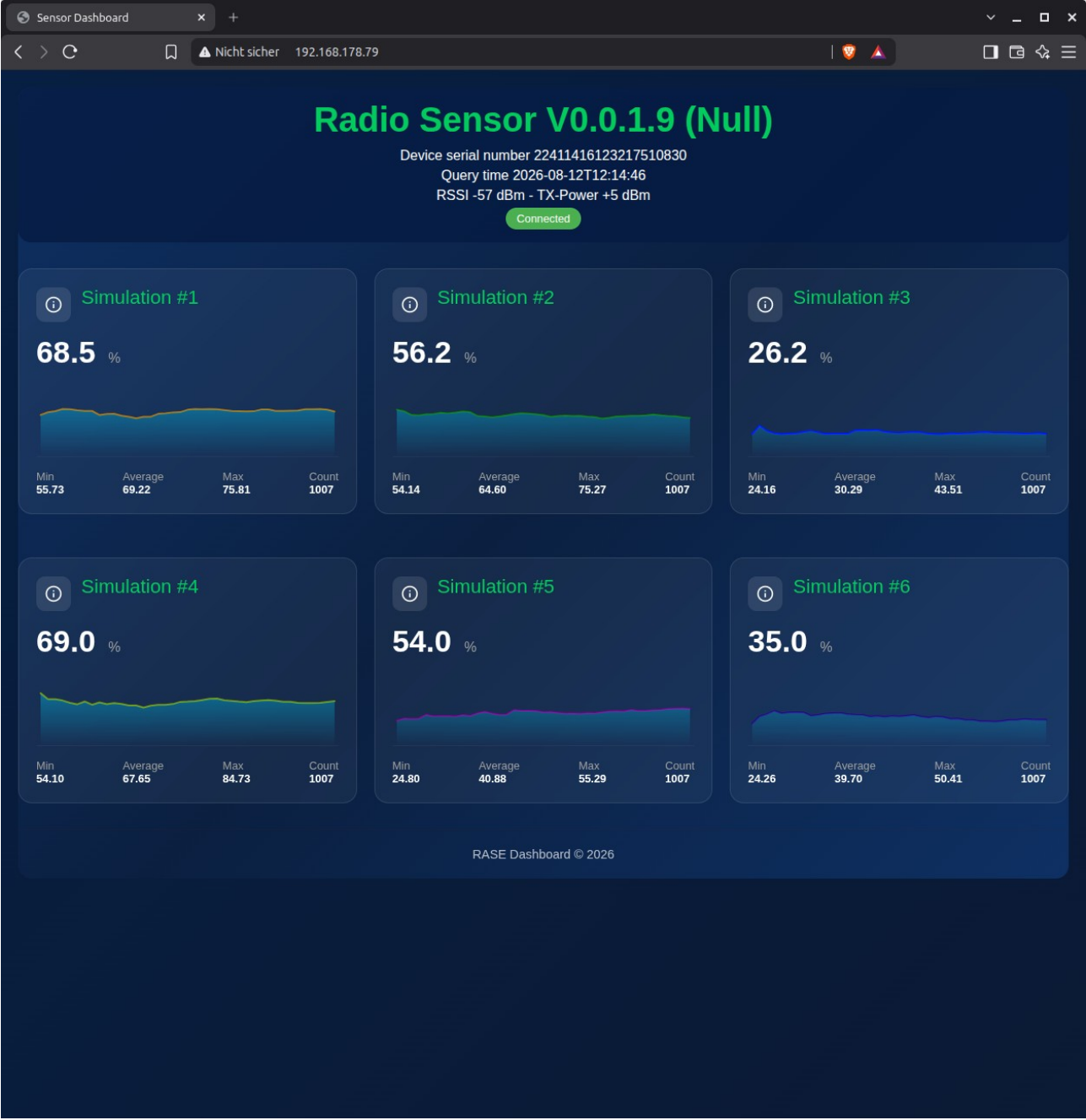
RESTful interface, endpoints over HTTP:

"/"	HTTP GET	Root website.
"/config"	HTTP GET	Configuration website.
"/update"	HTTP GET	Update website.
"/favicon.ico"	HTTP GET	Favicon request.
"/api/sensors"	HTTP GET	Query currently sensor status, returns JSON string.
"/api/id"	HTTP GET	Query device identification, returns JSON string.
"/api/config"	HTTP GET HTTP PUT	Query device configuration, return JSON string. Write device configuration, writes back JSON string.
"/api/reboot"	HTTP GET	Causes device reboot. Returns status JSON string.
"/api/update"	HTTP PUT	Receives new firmware image (binary).

Terminal interface, endpoints over TTY:

"/root"	Returns HTTP source of root website.
"/connect"	Configures WiFi connection, format: „/connect=Wifi-SSID:password“
"/broker"	Configures MQTT broker, format: /broker=IP-addr[:username[:password]]“
"/par"	Set a single config parameter. Allowed keys are all keys of config JSON string. Format: „/par=key:value“
"/api/sensors"	Query currently sensor status, returns JSON string.
"/api/id"	Query device identification, returns JSON string.
"/api/config"	Query device configuration, return JSON string.
„/api/config=“	Write device configuration, writes back JSON string. Parameter = config JSON.
"/api/reboot"	Forces an immediate device reboot.
"/api/initialize"	Forces a factory init. All config parameter are set to their defaults.

Root website layout (example):



Config website (example):

Rase Configuration Manager

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192.168.178.79/config

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RASE Configuration Manager

Manage device settings and save changes to the server

WiFi Settings

SSID

Password

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WiFi Channel

Channel 1

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MQTT Settings

MQTT Broker IP

192.168.178.21

MQTT Username

MQTT Password

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MQTT Enable

Disabled

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Display Settings

Display Layout

Value page

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Update website (example):



Sensor status JSON (example):

```
{
  "sensor": "Radio Sensor V0.0.1.9 (Null)",
  "device_serial_number": "22411416123217510830",
  "timestamp": "2026-08-12T12:01:52",
  "rssi": "-57 dBm",
  "tx_power": "+5 dBm",
  "channels": 6,
  "nodes": [
    { "sim_1": {
      "name": "Simulation #1",
      "shortcut": "X1",
      "unit": "%",
      "color": "#bb6a00",
      "bottom": 0.000,
      "top": 100.000,
      "value": 61.514,
      "min": 59.086,
      "max": 78.542,
      "average": 69.963,
      "flags": 33,
      "count": 1692,
      "history": [ 78.5, 65.3, 73.2, 74.2, 69.3, 71.0, 71.8, 72.6, 68.0, 70.5, 68.1, 69.0, 68.7, 73.2, 72.5, 73.5, 74.9, 74.5, 75.0,
67.7, 69.0, 69.6, 69.4, 67.2, 66.1, 66.9, 68.7, 68.3, 66.4, 65.0, 62.2 ] }
    },
    { "sim_2": {
      "name": "Simulation #2",
      "shortcut": "X2",
      "unit": "%",
      "color": "#126e00",
      "bottom": 0.000,
      "top": 100.000,
      "value": 34.783,
      "min": 15.922,
      "max": 42.184,
      "average": 31.211,
      "flags": 33,
      "count": 1692,
      "history": [ 16.4, 25.9, 31.7, 30.2, 26.1, 35.2, 35.6, 34.2, 34.9, 37.6, 40.0, 37.7, 34.9, 35.9, 32.7, 30.6, 32.7, 37.7, 40.8,
37.9, 35.4, 35.1, 35.7, 34.9, 29.8, 30.9, 34.5, 34.8, 35.8, 35.9, 36.6, 35.2, 31.2, 29.4, 29.7, 28.9, 28.1, 26.0, 25.4, 25.7, 26.9, 26.4,
25.8, 26.5, 28.3, 28.1, 28.1, 28.5, 29.3, 30.0, 31.5, 32.9, 32.4, 34.2 ] }
    }
  ]
}
```

Device identification JSON (example):

```
{
  "identification": "RASE",
  "manufacturer": "KKS-Elektronik",
  "product": "Radio Sensor",
  "device_serial_number": "04804824902214116000",
  "firmware_version": "0.0.1.9",
  "firmware_date": "2026-08-19",
  "chip_type": "ESP32-S3 Wroom",
  "sensor_head": "SHT3x",
  "head_serial_number": "0x1AEB1E21",
  "wifi_station_mac": "30:30:F9:16:8D:A0",
  "wifi_ap_mac": "30:30:F9:16:8D:A1",
  "bluetooth_mac": "30:30:F9:16:8D:A2",
  "ip_addr": "192.168.178.52",
  "partition-label": "app1",
  "partition-size": "1.5 MB",
  "flash-chip-id": "0x005E4018",
  "flash-chip-size": "8.0 MB",
  "spi-ram": "8192 KB",
  "heap-size-kb": "8520 KB",
  "rssi": "-42 dBm",
  "tx_power": "+5 dBm",
  "system_time": "2026-08-19T16:39:22"
}
```

Config JSON (example):

```
{
  "version": "0x20260710",
  "ssid": "YOUR SSID",
  "password": "YOUR PASSWORD",
  "wifi_channel": "1",
  "mqtt_broker": "192.168.178.21",
  "mqtt_username": "YOUR MQTT USERNAME",
  "mqtt_password": "YOUR MQTT PASSWORD",
  "mqtt_enable": "disabled",
  "display_layout": "Value page",
  "display_param": "0",
  "display_rotation": "0°",
  "display_timeout": "Never",
  "display_contrast": "10%",
  "sensor_type": "autoscan,autoscan,Null,SHT2x,SHT3x,HTU21d,ATHxx,HDC1080,BMx280",
  "led_intensity": "1%"
}
```

Reboot status JSON (example):

```
{
  "device_serial_number": "22411416123217510830",
  "reboot_time_stamp": "2026-08-12T12:06:15"
}
```